

Lab 4

Introduction to Arithmetic and Logical Instructions

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Arithmetic Instruction in 8086 Microprocessor

Arithmetic Instructions are the instructions which perform basic arithmetic operations such as addition, subtraction and a few more.

Following is the table showing the list of arithmetic instructions:

OPCODE	OPERAND	EXPLANATION	EXAMPLE
ADD	D, S	$D = D + S$	ADD AX, BX
ADC	D, S	$D = D + S + \text{prev. carry}$	ADC AX, BX
SUB	D, S	$D = D - S$	SUB AX, [SI]
SBB	D, S	$D = D - S - \text{prev. carry}$	SBB BX, 0050
MUL	register	$EAX = EAX * \text{register}$	MUL CX / MUL ECX
IMUL	register	performs signed multiplication	IMUL CX
DIV	register	$EAX = EAX / \text{register (8 bit). ; AL = quotient ; AH = remainder}$	DIV BL
DIV	16-bit register	$EAX / 16\text{-bit reg. ; AX = quotient ; DX = remainder}$	DIV CX
IDIV	16 bit register	performs signed division	IDIV BX
INC	D	$D = D + 1$	INC AX
DEC	D	$D = D - 1$	DEC [2050]
CWDE	none	convert word to doubleword	CWDE
NEG	D	$D = 2\text{'s compliment of } D$	NEG CX

Here D stands for destination and S stands for source.
D and S can either be register, data or memory address.

Logical Instruction in 8086 Microprocessor

Logical instructions are the instructions which perform basic logical operations such as AND, OR, etc.

Following is the table showing the list of logical instructions:

OPCODE	OPERAND	DESTINATION	EXAMPLE
AND	D, S	D = D AND S	AND AX, 0010
OR	D, S	D = D OR S	OR AX, BX
NOT	D	D = NOT of D	NOT AL
XOR	D, S	D = D XOR S	XOR AL, BL
TEST	D, S	performs bit-wise AND operation and affects the flag register without changing D	TEST [0250], 06
SHR	D, C	shifts each bit in D to the right C times and 0 is stored at MSB position	SHR AL, 04
SHL	D, C	shifts each bit in D to the left C times and 0 is stored at LSB position	SHL AX, BL
ROR	D, C	rotates all bits in D to the right C times	ROR BL, CL
ROL	R, C	rotates all bits in D to the left C times	ROL BX, 06
RCR	D, C	rotates all bits in D to the right along with carry flag C times	RCR BL, CL
RCL	R, C	rotates all bits in D to the left along with carry flag C times	RCL BX, 06

Here D stands for destination, S stands for source and C stands for count.

They can either be register, data or memory address.

Lab Work

Consider following data segment, where a, b, mask and x has already been initialized to memory. There are some unknown variables as well.

Make a project **my_calc** and write a program that will calculate all the missing values and store them into the dedicated variables.

For Report:

- Add your code (Readable code with appropriate comments for each arithmetic and logical instruction being used)
- After execution of the program, add screenshot of watch window which has every variable.
- Submit doc or pdf

.data

a	WORD	9	
b	WORD	5	
mask	WORD	30	
x	WORD	2	
sum	WORD	?	; sum = a+b
sub1	WORD	?	; sub1 = a-b
product	WORD	?	; product = a*b
quotient	WORD	?	; quotient = a/b
remainder	WORD	?	; remainder = a%b
a_minus	WORD	?	; a_minus = -a
b_plus1	WORD	?	; b_plus1 = b+1 (Don't use ADD)
a_and_mask	WORD	?	; a_and_mask = a & mask
b_or_mask	WORD	?	; b_or_mask = b mask
a_shr_x	WORD	?	; a_shr_x = shift right a by x times
b_shl_x	WORD	?	; b_shl_x = shift left b by x times
a_ror_x	WORD	?	; a_ror_x = rotate right a by x times
b_rcl_x	WORD	?	; b_rcl_x = rotate left with carry b by x times